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Common reporting errors in subgroup analysis: a comparison of interaction and stratified regression models

Chingkwong Tsoi[1] and Ran Sun[1*]

Abstract

Background In public health, epidemiology and medical research, it is considered a relatively effective practice to report treatment effect estimates from stratified regressions alongside significance test results from interaction regressions. However, since the two approaches differ in their theoretical foundations, statistical properties, interpretability, and conditions for appropriate application, combining them in reporting may raise methodological concerns. This study aims to evaluate and compare the effectiveness of these two methods in estimating heterogeneous treatment effects using Monte Carlo simulations.

Methods We conducted a series of Monte Carlo simulations to compare the performance of interaction regression and stratified regression across various scenarios. These scenarios varied in total sample size, the distribution of subgroup proportions, heterogeneity in the associations between covariates and the outcome variable across subgroups, and the degree of correlation among covariates. We simulated 10 covariates and used a generalised linear model to generate the outcome variable. The performance of both methods was evaluated using mean squared error (MSE), empirical coverage rate (ECR), and empirical statistical power (ESP).

Results The results demonstrated that stratified regression based on bootstrap resampling effectively controlled Type I error under large sample sizes and balanced group proportions. Interaction regression struggled to control Type I error when the coefficients of baseline characteristics differed across groups, and covariates were only weakly correlated. In other scenarios, the interaction regression outperforms stratified regression, primarily due to its relatively higher statistical power while maintaining acceptable Type I error rates. As an illustrative application, the two methods were applied to real-world data from the International Social Survey Programme (ISSP) to demonstrate how interaction regression and stratified regression may yield different conclusions regarding the association between fruit or vegetable consumption and body mass index.

Conclusion This study highlights the strengths and limitations of interaction regression and stratified regression in estimating heterogeneous treatment effects. Researchers should avoid adopting a hybrid strategy for reporting subgroup analysis results, such as presenting treatment effect estimates from stratified regressions and significance tests from interaction regressions, because the two methods may exhibit substantial differences in Type I error control under certain conditions.

*Correspondence:
Ran Sun
ransun@smail.nju.edu.cn

Full list of author information is available at the end of the article

Keywords Subgroup analyses, Interaction regression, Stratified regression, Heterogeneous treatment effects, Type I error control, Monte Carlo simulations

Background

Assessing heterogeneous treatment effects through subgroup analysis has become a widely used approach in public health, epidemiology and medical research [1]. This method helps researchers determine which groups derive greater or lesser benefit from the intervention. A typical example is the significantly reduced or even absent efficacy of cetuximab in metastatic colorectal cancer patients with mutant KRAS [2].

From a statistical modelling perspective, such analyses are often classified as effect modification or moderation analyses, which aim to assess whether the strength of a treatment effect varies systematically across subpopulations. The presence or absence of heterogeneous treatment effects depends on the specific research goal and the scale on which the treatment effect is evaluated [1, 3]. Methodologically, two primary estimation frameworks are commonly employed to evaluate heterogeneous treatment effects on a relative scale [4]. The first involves incorporating interaction terms in regression models (referred to as interaction regression) to test whether one or more variables modify the treatment effect [5]. The second approach is stratified regression, also known as subgroup-specific analyses, where treatment effects are estimated separately within predefined subgroups and then compared across subgroups. In subgroup analysis, the evaluation of heterogeneous treatment effects typically relies more heavily on the use of interaction terms rather than on stratified regression [6, 7]. Although both frameworks are widely used in practice, they differ in their theoretical foundations, statistical properties, interpretability, and conditions for appropriate application.

Most guidelines provide researchers with recommendations on which modelling approaches to choose, considering factors such as the type of model, the dimensionality of the covariate space, and the need to prevent overfitting [8–12]. Interaction regression is often favoured because of the simplicity of its inferential procedures. For most studies, reporting both the treatment effect estimates from stratified regression and the results of significance tests from interaction regression is regarded as a relatively effective practice [13]. For example, in a study on the association between an Alzheimer's disease biomarker and change in muscle strength, researchers treated sex as a moderator, first using stratified regression to assess changes in the direction of the association, and then applying interaction regression to test the statistical significance of the moderation effect [14]. However, this practice may lead to interpretive inconsistencies, as the effect estimates and their statistical significance are based on different model assumptions. Moreover, mixing inference across frameworks may inflate the Type I error rate [15].

There is growing concern that many published studies suffer from methodological and statistical flaws [16, 17], with increasing attention also being paid to the potential for p-hacking and selective reporting [18, 19]. In support of these concerns, previous studies have shown that a substantial proportion of published research fails to provide details on how subgroup analyses are implemented or how statistical inference is conducted [6, 20]. Moreover, some researchers may lack a clear understanding of how certain statistical software packages implement specific algorithms and inferential procedures [21]. This presents a substantial challenge for confirmatory subgroup analyses, as such analyses require maintaining strict control of the Type I error rate rather than aiming for high statistical power [22–24]. In particular, it remains unclear how stratified and interaction regression differ in performance under scenarios where subgroup proportions are imbalanced, covariates are correlated, or treatment effects vary in magnitude and direction across subgroups. To address this gap, this study uses Monte Carlo simulation to evaluate the robustness and statistical power of these two methods and to provide practical recommendations for confirmatory research and data analysis. This study aims to quantify and compare the statistical performance of commonly used modelling strategies under varying data characteristics by evaluating stratified regression and a commonly employed interaction regression approach, in which only a single interaction term of primary interest is included while other potential interactions are omitted.

Methods

Overview

This chapter describes the study design for comparing the performance of interaction regression and stratified regression using Monte Carlo simulations. Section 2.2 to 2.3 provide a detailed explanation of the methods and procedures used to generate all variables. Section 2.4 to 2.5 provide the parameter settings for numerical simulations and the overall Monte Carlo simulation design. Section 2.6 outlines the performance evaluation metrics used to evaluate model performance.

Generation of covariates

We simulated 10 covariates for each sample. Specifically, we first generated a latent

10-dimensional continuous covariate vector $X^* = (X_1^*, \ldots, X_{10}^*)^{\mathrm{T}} \in \mathbb{R}^{10}$, where X_1^* represents the latent treatment variable and X_2^* *to* X_{10}^* denote latent baseline characteristics. The latent covariates were independently and identically distributed across observations and jointly followed a multivariate normal distribution with a zero mean vector and a symmetric positive definite covariance matrix $\Sigma \in \mathbb{S}_{++}^{10}$. To better reflect common empirical practice, the observed exposure variable X_1 and the first four baseline covariates X_2 to X_5 were constructed as binary indicators by dichotomising their corresponding latent continuous variables at the median. Specifically, $X_j = \mathbb{I}(X_j^* \geq \mathrm{median}(X_j^*))$ for $j = 1, \ldots, 5$. The remaining covariates X_6 to X_{10} were retained in their original continuous form. This specification of covariates was primarily motivated by its applicability in fields such as epidemiology and medicine, where the exposure variable is often categorical, representing two or more types of interventions, such as medications or treatment regimens. In addition, covariates in these fields typically include a combination of both categorical and continuous variables.

Generation of outcome variable

We simulated an outcome variable for each sample based on the generalised linear model (GLM). For each sample, let the covariate vector be $\boldsymbol{X}$, the random error be $\boldsymbol{\varepsilon}$, and the outcome vector be $\boldsymbol{y}$. The samples in the study are divided into two subgroups (A and B), with the outcome variable for each subgroup generated by different linear models:

$$\begin{array}{ll} \text{Model} \quad \text{A}: & \mathbf{y} = \mathbf{X}\alpha + \alpha_0 + \varepsilon \\ \text{Model} \quad \text{B}: & \mathbf{y} = \mathbf{X}\beta + \beta_0 + \varepsilon \end{array} \tag{1}$$

where coefficient vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ represent regression slopes, and $\boldsymbol{\alpha}_0$ and $\boldsymbol{\beta}_0$ represent intercepts. We introduced a binary indicator variable $\boldsymbol{D}$ (with value 0 for subgroup A and 1 for subgroup B) to integrate the two models into a unified equation:

$$\begin{aligned} \mathbf{y} &= (1-\mathbf{D})\mathbf{X}\alpha + \mathbf{D}\mathbf{X}\beta + (1-\mathbf{D})\alpha_0 + \mathbf{D}\beta_0 + \varepsilon \\ &= \mathbf{X}\alpha + \mathbf{D}\mathbf{X}\gamma + \alpha_0 + \mathbf{D}\gamma_0 + \varepsilon \end{aligned} \tag{2}$$

where the coefficient vector $\boldsymbol{\gamma} = \boldsymbol{\beta} - \boldsymbol{\alpha}$ denotes the difference in slopes between the two subgroups, and $\boldsymbol{\gamma}_0 = \boldsymbol{\beta}_0 - \boldsymbol{\alpha}_0$ denotes the difference in intercepts.

Statistical analyses of simulated datasets

We conducted simulation scenarios with different configurations. First, we considered two sample size levels: small sample ($N = 250$) and large sample ($N = 1000$). Second, we defined two subgroups, setting the proportion of subgroup A in the total sample at five levels (10%, 30%, 50%, 70%, and 90%), with subgroup B corresponding to the remaining proportion. Third, the difference in the regression coefficient of the treatment variable X_1 between the two subgroups (γ_1) was set to 0 and 0.5. In this case, a value of $\gamma_1 = 0$ indicates that the treatment effect is homogeneous, while a nonzero value suggests that the treatment effect is heterogeneous. Fourth, the differences in the regression coefficients of baseline characteristics (X_2 to X_{10}) between the two subgroups (γ_2 to γ_{10}) were set to $\gamma = 0$ and 0.5. Finally, the Pearson correlation coefficients between all covariates were set to $r_{ij} = 0$ and $|r_{ij}| < 0.2$, where $i \neq j$ and $i, j \in \{1, \ldots, 10\}$. In the $|r_{ij}| < 0.2$ condition, each pairwise correlation coefficient was randomly drawn from a uniform distribution $U(-0.2, 0.2)$.

The following parameter settings were applied consistently across all simulations. First, the parameters were set as follows: α_1 to α_{10} were fixed at 0.5, α_0 was set to 0, and γ_0 was set to 1. Second, all covariates were standardised to have a mean of 0 and a variance of 1. Third, the subgroup binary indicator remained independent of all covariates. Fourth, the random error term was assumed to follow a normal distribution $N(0,10)$. Finally, each configuration was simulated 3000 times.

To better reflect empirical research settings, the simulation design allows for heterogeneous effects of baseline characteristics across subgroups and incorporates correlation among covariates. It is evident that when $\gamma \neq 0$, failing to include the interaction between covariates and subgroup indicators in the regression model can result in omitted variable bias. In real-world applications, baseline characteristics often influence outcomes differently across population subgroups, and covariates are rarely independent due to shared underlying factors. Ignoring such heterogeneity and correlation may therefore lead to overly optimistic assessments of model performance.

Design of the Monte Carlo simulations

Using the previously described procedure, we generated the required variables for each simulation scenario. For each simulated dataset, both interaction regression and stratified regression models were estimated. In the interaction regression analysis, an ordinary least squares (OLS) regression was fitted to the full sample to estimate the coefficients of all covariates, the subgroup indicator, and the interaction term between the subgroup indicator and the treatment variable. A 95% confidence interval for the interaction term was then calculated. Although it is possible to include interaction terms between the subgroup indicator and all baseline covariates (i.e., the full interaction model), we did not include this scenario in our simulation since this method is rarely used in practice, and while its point estimates are numerically identical to those from stratified regression, their standard errors differ.

In the stratified regression analysis, we first separated the sample by subgroup and conducted independent linear regressions within subgroups A and B, directly obtaining point estimates of treatment effects for each subgroup. Subsequently, the uncertainty of the heterogeneous treatment effect was estimated using a bootstrap approach to generate 500 resampled datasets. For each bootstrap sample, we repeated the stratified regression process, calculated the difference in the regression coefficient of the treatment variable between the two subgroups, and constructed a percentile-based 95% confidence interval for the heterogeneous treatment effect based on the distribution from these 500 repetitions.

Performance measures

The performance of the treatment effect heterogeneity estimation approaches obtained from both the interaction regression and the stratified regression was evaluated using the mean squared error (MSE), the empirical coverage rate (ECR), and the empirical statistical power (ESP). MSE is defined as the average squared difference between the estimated and true parameter values, i.e., $MSE = E[(\widehat{\gamma}_1 - \gamma_1)^2]$, with lower values indicating higher estimation accuracy. ECR is defined as the proportion of simulated samples in which the 95% confidence interval of the estimate contains the true parameter value. Under the duality between confidence intervals and hypothesis testing, the Type I error rate corresponds to the proportion of simulations in which the $(1 - \alpha)$ confidence interval for γ_1 fails to include its true value. Consequently, the closer the ECR is to the nominal 95% level, the better the constructed confidence intervals maintain nominal Type I error control. ESP is defined as the proportion of simulated samples in which the null hypothesis H_0: $\gamma_1 = 0$ was rejected at a significance level of $\alpha = 0.05$. In simulation scenarios where $\gamma_1 = 0$, statistical power was not assessed.

The simulation design described above was conducted within the framework of general linear models and has not been extended to other types of outcome variables. Additional File 1 provides additional details on the extension of this simulation procedure to binary outcomes, including the methodology and corresponding results. All simulations and statistical analyses were conducted in R (version 4.4.0). The R code used for the Monte Carlo simulations is provided in Additional File 2.

Simulation results

Figure 1 presents the MSE across all the simulation scenarios. As expected, in all configurations, MSE decreased for both methods as the total sample size increased or as the distribution of subgroup proportions became more balanced. In most scenarios, especially when subgroup proportions were relatively balanced, the estimation precision of both methods was very similar, as reflected by nearly identical MSE values. In addition, whether treatment effects were homogeneous or heterogeneous, the overall differences in MSE between the two approaches were negligible. Similar patterns were observed for binary outcomes, while in scenarios with small sample sizes and unbalanced subgroup proportions, stratified regression exhibited substantially higher MSE than interaction regression (Additional File 1).

The ECR values under different simulation conditions are displayed in Fig. 2. For the stratified regression, it performed well in most configurations, with its ECR within 1 percentage point of the nominal 95% level, indicating generally reliable control of Type I error. However, when the sample size was small and subgroup proportions were highly imbalanced, the stratified regression exhibited substantial deviation in Type I error. In contrast, interaction regression maintained robust ECR performance in this scenario. When baseline characteristic coefficients differed between subgroups and covariates were correlated, interaction regression exhibited substantial deviations from the nominal level. This result is expected under model misspecification, as omitting the true interaction terms that are correlated with the included interaction term induces omitted variable bias. Overall, when the sample size was large ($N = 1000$), the ECR deviated more from the nominal level as subgroup proportions became more balanced. Conversely, when the sample size was small ($N = 250$), the deviation decreased as proportions became more balanced. However, this pattern of deviation was not observed for interaction regression in the simulations with binary outcomes (Additional File 1).

The ESP values under different simulation conditions are shown in Fig. 3. Across all simulation configurations, the ESP for both the stratified regression and the interaction regression increased as the subgroup proportions became more balanced or as the sample size increased. Under conditions of high imbalance in subgroup proportions and small sample sizes, the stratified regression had considerably lower statistical power compared to the interaction regression. This finding suggests that in such scenarios, the stratified regression achieves higher coverage rates at the expense of statistical power, resulting in overly conservative interval estimates. When the sample size was 1000, the performance gap in ESP between the two approaches narrowed. However, when baseline characteristic coefficients differed between subgroups and the sample size was large, the ESP of interaction regression was lower than that of stratified regression in most scenarios in the simulations with binary outcomes (Additional File 1).

The results above indicate that stratified regression tends to exhibit reduced statistical performance when

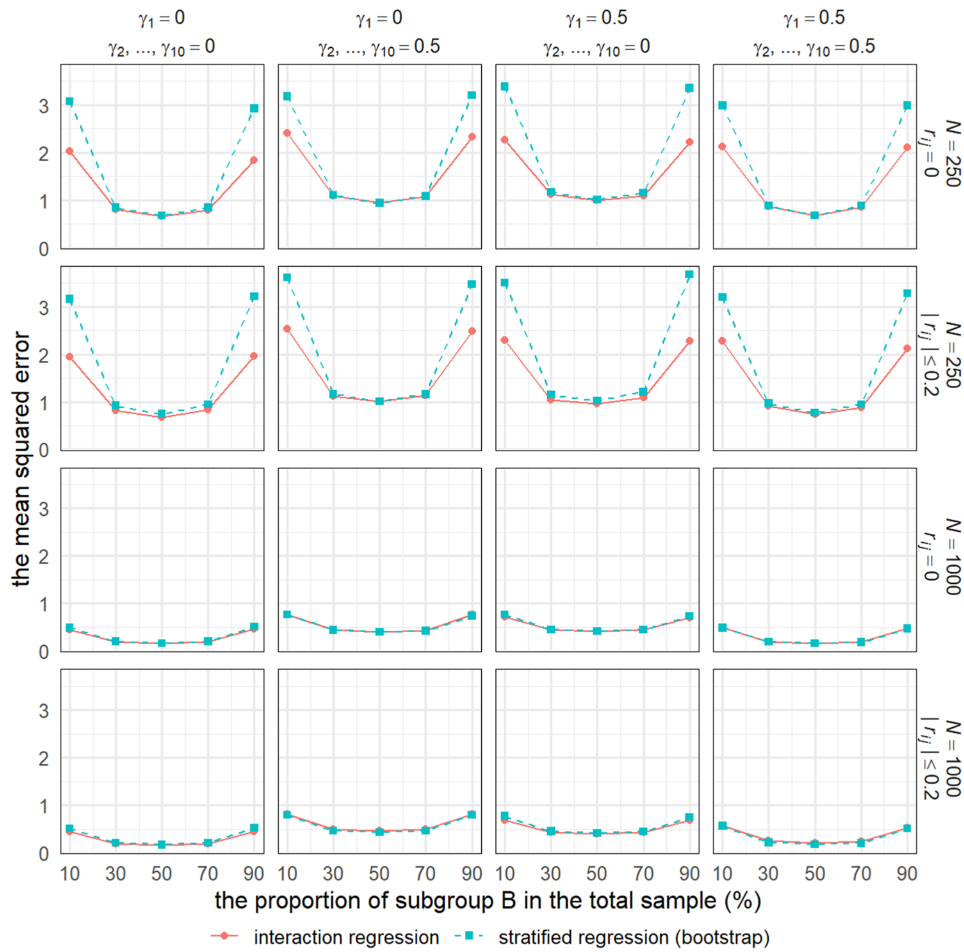

Fig. 1 The mean squared error of interaction regression and stratified regression was obtained in each scenario of the Monte Carlo simulations. N denotes the total sample size for each simulated dataset; γ_1 represents the heterogeneous treatment effect; γ_2 to γ_{10} indicate differences in the regression coefficients of control variables across subgroups; r refers to the Pearson correlation coefficient between each pair of covariates

the total sample size is small and subgroup proportions are highly imbalanced, primarily due to overfitting when the sample size within a subgroup is small relative to the number of parameters to be estimated. For example, in scenarios where subgroup A contains 25 observations and subgroup B contains 225 observations, stratified regression estimates the same number of coefficients in each subgroup, leading to limited degrees of freedom and increased estimation instability in the smaller subgroup. To further investigate this issue, additional simulation settings with fewer covariates are presented in Additional File 3. Specifically, the outcome was generated using only four covariates, consisting of one exposure variable, two binary covariates, and one continuous covariate. Under these reduced parameter settings, the deviation of the ECR in stratified regression is substantially attenuated, even when subgroup proportions remain highly imbalanced. These results suggest that the performance of stratified regression is more strongly constrained by overfitting arising from small subgroup sample sizes relative to model complexity.

Applied example

Data and statistical analysis

To illustrate the practical differences between the two approaches, we used data from the Health and Health Care module of the 2021 wave of the International Social Survey Programme (ISSP) as an empirical example. The ISSP is a recurring cross-national collaborative project that conducts standardised questionnaire surveys on specific topics. Sampling within the ISSP was independently implemented by participating national research organisations, following methodological guidelines established by the ISSP Methodology Committee. In general, countries are required to employ probability-based sampling methods, such as stratified random sampling or multistage cluster sampling, to draw nationally representative samples of individuals aged 18 and above [25]. The

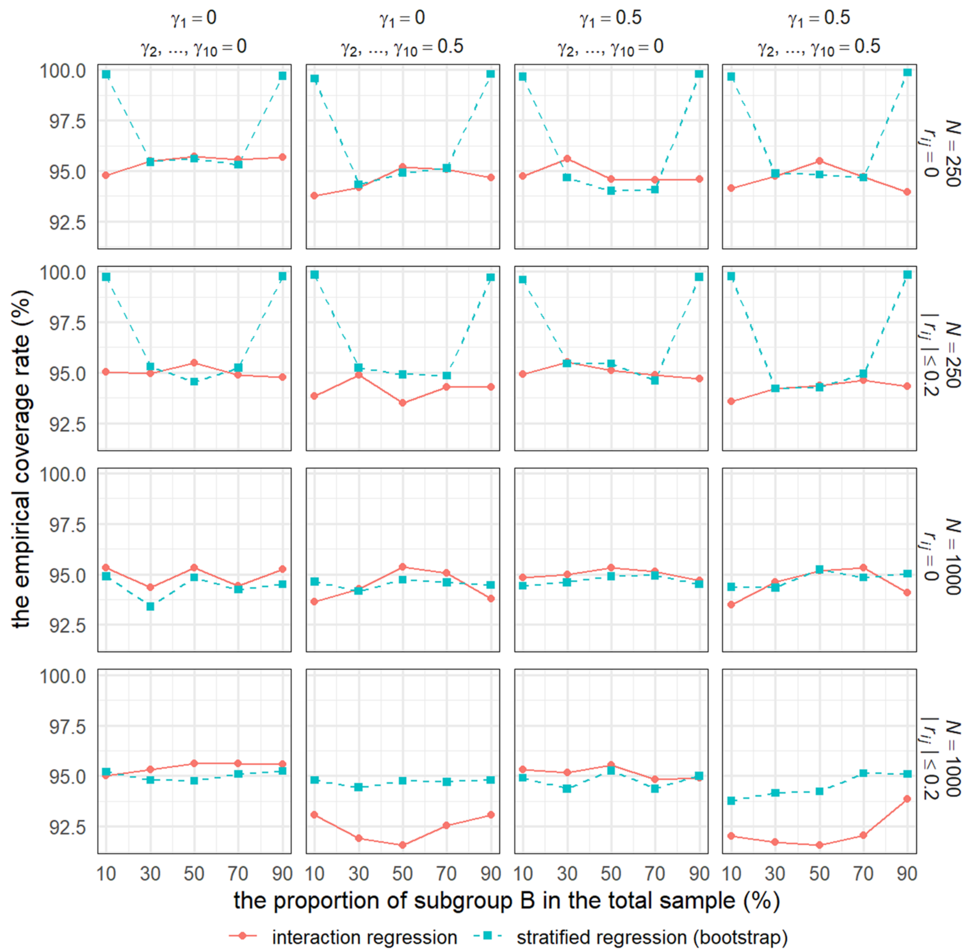

Fig. 2 The empirical coverage rate of interaction regression and stratified regression was obtained in each scenario of the Monte Carlo simulations. N denotes the total sample size for each simulated dataset; γ_1 represents the heterogeneous treatment effect; γ_2 to γ_{10} indicate differences in the regression coefficients of control variables across subgroups; r refers to the Pearson correlation coefficient between each pair of covariates

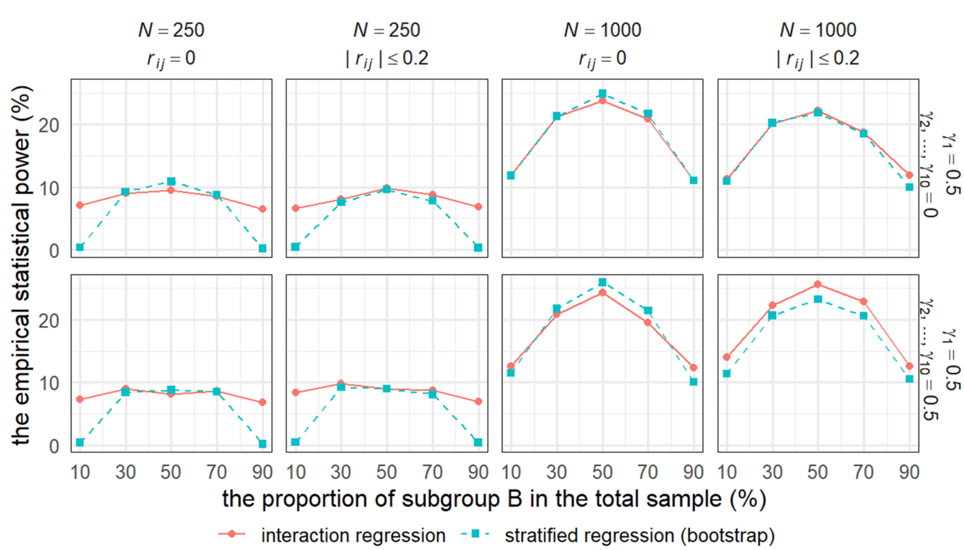

Fig. 3 The statistical power of interaction regression and stratified regression was obtained in each scenario of the Monte Carlo simulations. N denotes the total sample size for each simulated dataset; γ_1 represents the heterogeneous treatment effect; γ_2 to γ_{10} indicate differences in the regression coefficients of control variables across subgroups; r refers to the Pearson correlation coefficient between each pair of covariates

anonymised microdata are publicly available from the official ISSP data archive.

As a concrete example, we analysed whether the relationship between fruit or vegetable consumption frequency and body mass index (BMI) in adults aged 20 to 64 years is moderated by sex or age. Information on dietary behaviour was derived from the survey question "How often do you eat fresh fruit or vegetables?" Based on respondents' self-reported frequency, we constructed a binary indicator denoting whether individuals reported consuming fruits or vegetables daily. BMI was calculated as weight in kilograms divided by height in meters squared, using self-reported height and weight. Baseline characteristics included age, sex, years of education, employment status, physical activity frequency, smoking status, alcohol consumption frequency, presence of a chronic illness, and residential location. Interaction regression (including only the interaction between the exposure variable and subgroup indicators), full interaction regression (including interactions between all covariates and subgroup indicators), and stratified analysis were implemented, both using OLS estimation. These results are included primarily to demonstrate how interaction regression can identify effect heterogeneity, and should not be interpreted as evidence of clinically meaningful sex-specific differences.

Results

Table 1 presents the estimated moderating effects of fruit or vegetable consumption frequency on BMI across subgroups defined by sex and years of education, derived from both interaction regression and stratified regression. All models were adjusted for baseline characteristics to reduce potential confounding and were weighted using survey design weights. The effective sample size was 23,951, including 11,251 males and 12,700 females. The distribution across education groups was as follows: 2917 individuals with 0–9 years of education, 7343 with 10–12 years of education, and 13,691 with 13 or more years of education.

In the analysis of whether the relationship between fruit or vegetable consumption frequency and BMI varies by sex, the interaction term in the interaction regression was statistically significant, with a coefficient of 0.371 ($t = 3.222$, $p = 1.3\text{e}{-3}$). The likelihood ratio test comparing nested models also confirmed that the inclusion of the interaction term significantly improved the model's goodness of fit (LR = 10.388, $p = 1.3\text{e}{-3}$). In addition, the Wald test of linear constraints on the interaction coefficient provided consistent evidence (W = 10.383, $p = 1.3\text{e}{-3}$). Although the full interaction regression produced a similar effect estimate ($\beta = 0.283$, $t = 2.403$, $p = 0.016$), the coefficient was smaller and less statistically significant. In contrast, the stratified regression failed to detect a statistically significant interaction effect at the 5% level. The stratified estimates indicated that the main effect of fruit or vegetable consumption frequency was 0.284 (95% CI: − 0.042, 0.603) higher among males compared to females. A percentile-based 95% confidence interval constructed from 1000 bootstrap replications included zero, and therefore, the null hypothesis of no interaction was not rejected in a two-sided test. The bootstrap-assisted Wald test yielded consistent inference. This method entails generating 1000 bootstrap samples and computing the empirical distribution of the statistics to estimate their variance or covariance matrix. Based on the bootstrap-estimated covariance matrix and the point estimates obtained from the original sample, the Wald statistic is constructed as a quadratic form of the parameter deviation, and statistical significance is determined using the critical values from the chi-square distribution [26].

Additional subgroup analyses across multiple categories were conducted to provide further insights. The interaction regression identified two statistically significant interaction effects in subgroups defined by education status: a coefficient of −0.379 ($t = -1.994$, $p = 0.046$) for the 10–12 years group relative to the 0–9 years group, and −0.809 ($t = -4.590$, $p = 4.5\text{e}{-6}$) for the 13 + years group relative to the 0–9 years group. Likelihood ratio tests and Wald tests supported these findings. In the full interaction regression, the coefficient of the 10–12 years group relative to the 0–9 years group was −0.384 ($t = -1.952$, $p = 0.050$), which did not reach conventional levels of statistical significance. In stratified regression, the estimated main effect of fruit or vegetable consumption frequency was 0.384 (95% CI: −0.935, 0.154) lower in the 10–12 years subgroup and 0.764 (95% CI: −1.259, −0.254) lower in the 13 + years subgroup compared to the 0–9 years subgroup. The percentile-based 95% confidence interval indicated that only the difference between the 13 + years subgroup and the 0–9 years subgroup reached statistical significance at the 5% level. The bootstrap-assisted Wald test further supported a statistically significant interaction between education level and fruit or vegetable consumption frequency, suggesting that the impact of dietary intake on BMI varies by educational attainment.

The conclusions derived from the interaction regression and stratified regression models may differ due to several contributing factors. First, fruit or vegetable consumption frequency may be subject to selection bias. When both the exposure and subgroup membership are jointly influenced by an unobserved confounder, self-selection may induce a spurious association between the exposure and the subgroup indicator, thereby leading to biased estimates of effect modification in interaction regression. At the same time, stratified regression is also

Table 1 Subgroup analyses based on interaction and stratified regression models, along with corresponding statistical tests

	Coefficient	**95% CI**	**t (*p*-value)**	**LR (*p*-value)**	**F (*p*-value)**	**W (*p*-value)**
Subgroups: Sex						
Interaction regression						
Female	-0.729	-0.886, -0.572				
Male	-0.358	-0.524, -0.192				
Diff: Male - Female	0.371	0.145, 0.596	3.222 (1.3e-3)			
Diff: Overall				10.388 (1.3e-3)	10.383 (1.3e-3)	
Full interaction regression						
Female	-0.679	-0.838, -0.519				
Male	-0.395	-0.563, -0.228				
Diff: Male - Female	0.283	0.052, 0.515	2.403 (0.016)			
Diff: Overall				82.973 (< 0.001)	5.775 (0.016)	
Stratified regression						
Female	-0.679	-0.845, -0.513				
Male	-0.395	-0.554, -0.236				
Diff: Male - Female	0.284	-0.042, 0.603				
Diff: Overall						2.904 (0.088)
Subgroups: Years of education						
Interaction regression						
0–9	0.022	-0.291, 0.335				
10–12	-0.356	-0.562, -0.151				
13+	-0.787	-0.938, -0.636				
Diff: (10–12) - (0–9)	-0.379	-0.751, -0.006	-1.994 (0.046)			
Diff: (13+) - (0–9)	-0.809	-1.154, -0.464	-4.590 (4.5e-6)			
Diff: Overall				26.200 (3.1e-7)	13.098 (2.1e-6)	
Full interaction regression						
0–9	0.009					
10–12	-0.375	-0.585, -0.164				
13+	-0.754	-0.907, -0.601				
Diff: (10–12) - (0–9)	-0.384	-0.770, 0.002	-1.952 (0.050)			
Diff: (13+) - (0–9)	-0.763	-1.121, -0.406	-4.189 (2.8e-5)			
Diff: Overall				112.095 (< 0.001)	10.514 (2.7e-5)	
Stratified regression						
0–9	0.009	-0.338, 0.357				
10–12	-0.375	-0.595, -0.154				
13+	-0.754	-0.901, -0.608				
Diff: (10–12) - (0–9)	-0.384	-0.935, 0.154				
Diff: (13+) - (0–9)	-0.764	-1.259, -0.254				
Diff: Overall						10.790 (0.005)

Coefficients are derived from OLS estimates, including main effects, sums of main and interaction effects, or differences between main effects. *t*-values are test statistics for the significance of individual regression coefficients. LR values are likelihood ratio test statistics for comparing nested models. F values are Wald test statistics used to jointly test linear constraints on one or more interaction terms. W values are derived from Wald tests based on bootstrap-estimated covariance matrices. The full interaction regression is defined as one that includes interaction terms between the subgroup indicator and all baseline covariates, in addition to the main effects

subject to sample selection bias, which can result in effect estimates that deviate from the true values. Second, the distribution of BMI deviated from normality. Although ordinary least squares estimation is consistent under fairly weak distributional assumptions, deviations from normality can affect finite-sample performance, particularly the accuracy of standard errors and confidence intervals, which may further contribute to differences in inference between the two modelling approaches.

Discussion

Our findings confirm that the interaction regression offers advantages in terms of overall statistical power, although previous studies have shown that interaction tests generally exhibit low sensitivity and have limited power to detect significant subgroup effects [22]. However, a major limitation of the interaction regression lies in its poor control of Type I error under conditions in which the coefficients of baseline characteristics differ between subgroups and the covariates are correlated.

This limitation became more pronounced for normally distributed outcome variables as the distribution of subgroup proportions became more balanced and the total sample size increased, whereas this pattern was not observed for binary outcome variables. The reduced estimation performance of the interaction regression in such scenarios primarily stems from omitted variable bias. Simulation scenarios involving heterogeneous effects of baseline characteristics and correlated covariates more closely resemble real-world statistical applications compared to other simulation conditions.

Our simulations further indicate that the stratified regression should be avoided under conditions of small sample sizes and extreme imbalance in subgroup proportions, as it performs poorly in terms of Type I error control and statistical power, potentially due to an elevated risk of overfitting. In scenarios where the coefficients of baseline characteristics differ systematically between subgroups and the covariates are correlated, the stratified regression significantly outperforms the interaction regression in controlling Type I error when the sample size is large. The stratified regression maintains adequate estimation precision and statistical power while ensuring reliable Type I error control in most configurations. These results suggest that the effectiveness of the bootstrap approach in subgroup analysis is largely constrained by the sample size of the smallest subgroup.

These performance differences are closely tied to their respective theoretical foundations. A major advantage of interaction regression is that it leverages the full sample, allows for adjustment of potential confounders, and generally offers higher statistical power compared with stratified approaches. It is particularly suitable when moderators are continuous [13], as it avoids categorisation-induced information loss. However, in high-dimensional or nonlinear settings, model misspecification may reduce estimation performance [27], and nonlinearity may be misinterpreted as interaction [28]. In stratified regression, statistically significant treatment effects within separate subgroups do not necessarily imply heterogeneity [29], differences across subgroups can be formally assessed using procedures such as the bootstrap [30, 31]. However, stratification reduces the effective sample size within each stratum, which in turn lowers statistical power, especially when subgroup sizes are small or the number of subgroups is large. This simulation study is based on ordinary least squares estimation. Violations of homoscedasticity and error independence may affect the validity of statistical inference in both interaction and stratified regression models. In empirical applications, these issues can be addressed by employing cluster robust standard errors. Alternatively, multilevel linear models may be used to explicitly model data dependencies and thereby mitigate these limitations.

The complex relationships commonly observed between covariates and outcomes in real-world data may exacerbate the divergence between simulation scenarios and actual empirical settings [32]. In addition to randomised controlled trials, subgroup analyses are frequently conducted in observational or cohort studies, which are more vulnerable to endogeneity, such as sample selection bias [33] and self-selection bias [34]. Previous research has also demonstrated that failing to account for differences in subgroup representativeness [35] or for interactions between moderators and confounders in the estimation of heterogeneous treatment effects can lead to systematic bias [5]. In practice, propensity score methods are commonly regarded as quasi-experimental approaches for addressing endogeneity in observational research [36], and they have also been extensively used in subgroup analyses to account for baseline differences in covariate distributions [37]. However, existing literature has emphasised that while propensity scores can alleviate issues related to functional form misspecification, they are insufficient to correct for sample selection, self-selection, or omitted variable bias [38, 39].

This study has several limitations. First, the Monte Carlo simulations assumed that both the outcome variable and the covariates followed a normal distribution. This assumption limits the generalisability of the findings to scenarios involving non-normally distributed outcome variables [27]. For example, there is an ongoing debate regarding the interpretation of interaction effects in generalised linear models, particularly concerning distinguishing between multiplicative and marginal interaction effects on both the probability and odds scales [40]. Previous research has also demonstrated that the existence of interactions depends on the type of effect measures used to quantify the relationship between treatments and outcomes [41]. Therefore, future research should further investigate the performance difference between interaction models and subgroup regression models under various types of outcome variables and covariate distributions. Such investigations would contribute to a better understanding of the sensitivity of these methods to distributional assumptions.

Second, this simulation study assumes that the researcher correctly specifies linear relationships between all observed independent variables and the outcome. In empirical applications, however, the true functional forms are often unknown, which may lead to model misspecification and biased estimates. To mitigate this limitation, complementary modelling strategies such as generalised additive models and propensity score methods may be considered. These methods provide greater flexibility in capturing complex relationships while retaining a reasonable level of interpretability.

Third, the subgroup definitions in this study were restricted to binary splits, whereas in practical research, the number of subgroups may be larger. As the number of subgroups increases, the distribution of the sample sizes across these groups becomes more complex, which may influence model performance in terms of statistical power and error control [42]. Future research should systematically evaluate the applicability and robustness of different modelling strategies under more complex subgroup structures. Beyond statistical properties, practical considerations are central to method selection in applied subgroup analyses. Bootstrap-based stratified regression can be computationally intensive, particularly in large samples or when many subgroups or covariates are considered, which may limit its feasibility in routine empirical work. By contrast, interaction regression is straightforward to implement using standard regression software and provides a coherent and familiar framework for hypothesis testing, making it more accessible to applied researchers. Moreover, although our simulations detect differences in statistical efficiency between the two approaches, these differences are often small and may have limited practical relevance in typical observational settings, especially when sample sizes are sufficiently large.

Fourth, this simulation study employed only two commonly used approaches to examine subgroup differences. On the one hand, subgroup differences in interaction analyses were assessed based on the regression coefficients of the interaction terms and their statistical significance. Generally, quantitative interactions are regarded as statistically more credible than qualitative ones, but our study focused exclusively on detecting the former [43]. The significance of interaction effects can also be evaluated by comparing the goodness-of-fit between nested models or by conducting a joint significance test for linear constraints on multiple interaction terms [44]. Besides analysing interactions using fixed effects, cross-random effects models can incorporate interaction effects into hierarchical residual components for analysis [45]. On the other hand, in addition to using bootstrap approaches to assess differences between subgroups in regression coefficients and their statistical significance, alternative approaches such as the permutation test and the Chow test are widely used [46–48]. The choice of method may influence the study's conclusions.

Conclusion

This study employed Monte Carlo simulations to evaluate the performance of interaction regression and stratified regression in estimating heterogeneous treatment effects. The results summarise the mean squared error, Type I error control, and statistical power of both methods under different simulated scenarios. The findings indicate that the stratified regression performs well in controlling Type I error under large sample conditions. When the coefficients of baseline characteristics differ across groups and the covariates are correlated, the bootstrap approach demonstrates robustness to imbalanced subgroup proportions. Under the same conditions, the interaction regression not only fails to adequately control the Type I error rate but also exhibits sensitivity to the distribution of subgroup proportions. In other scenarios, the interaction regression slightly outperforms stratified regression, primarily due to its relatively higher statistical power while maintaining acceptable Type I error rates. Researchers should avoid adopting a hybrid strategy for reporting subgroup analysis results, such as presenting treatment effect estimates from stratified regressions and significance tests from interaction regressions, because the two methods may exhibit substantial differences in Type I error control under certain conditions.

Abbreviations

KRAS	Kirsten Rat Sarcoma Viral Oncogene Homolog
ISSP	International Social Survey Programme
OLS	Ordinary Least Squares
MSE	Mean Squared Error
ECR	Empirical Coverage Rate
ESP	Empirical Statistical Power
BMI	Body Mass Index
GLM	Generalised Linear Model

Supplementary Information

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Additional File 1.

Additional File 2.

Additional File 3.

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Not applicable.

Authors' contributions
CT designed the study, analysed and interpreted the data, and critically revised the manuscript. RS collected, analysed and interpreted the data, and drafted and revised the manuscript.

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Data availability
The datasets generated and analysed during the current study are available in the ISSP data repository (https://www.gesis.org/en/issp/data-and-documentation/health-and-health-care/2021).

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]School of Social and Behavioral Sciences, Nanjing University, Nanjing, Jiangsu 210023, China

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